Clinical Decision Support System for Disease Risk Assessment

Progress Report - 5

Submitted By: Akiful Mohaimin (300369823)

Course: CSIS 4495 – Applied Research Project

Section: 03

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1. Work Date / Hours Log

Date	Hours	Description of Work Done
Mar 20, 2026	3.5	Assisted in redesigning the dashboard layout and improving UI structure in sampleapp.
Mar 21, 2026	4.0	Implemented comparison with past reports and data handling for trend tracking.
Mar 22, 2026	3.5	Developed visualization support and integrated trend chart data with UI.
Mar 23, 2026	4.0	Improved input sections (symptoms, patient data) and fixed UI rendering issues.
Mar 23, 2026	3.0	Collaborated on debugging UI, chart errors, and ensuring smooth integration across features.

2. Summary Description of Work Done During This Reporting Period

During this reporting period, I contributed to enhancing the functionality and interactivity of the frontend within sampleapp.py. The work focused on implementing features that improve how users interpret and interact with their data.

Key contributions included developing the comparison with past reports feature, enabling users to track changes in indicators over time. I also supported the integration of trend visualizations by ensuring proper data handling and compatibility with the UI.

Additionally, improvements were made to patient input sections, including symptoms and profile fields, making them more structured and user-friendly. Debugging efforts were also carried out to resolve UI inconsistencies and ensure smooth integration of charts and interactive features.

3. Repo Check-In of Implementation Completed

Updated File

- sampleapp.py

Work Done

Enhancements were made to sampleapp.py to improve interactivity and data handling within the frontend. This included implementing report comparison features, supporting trend visualization integration, refining input sections, and resolving UI-related issues. These updates ensured a smoother and more functional user experience.

4. AI Use Section

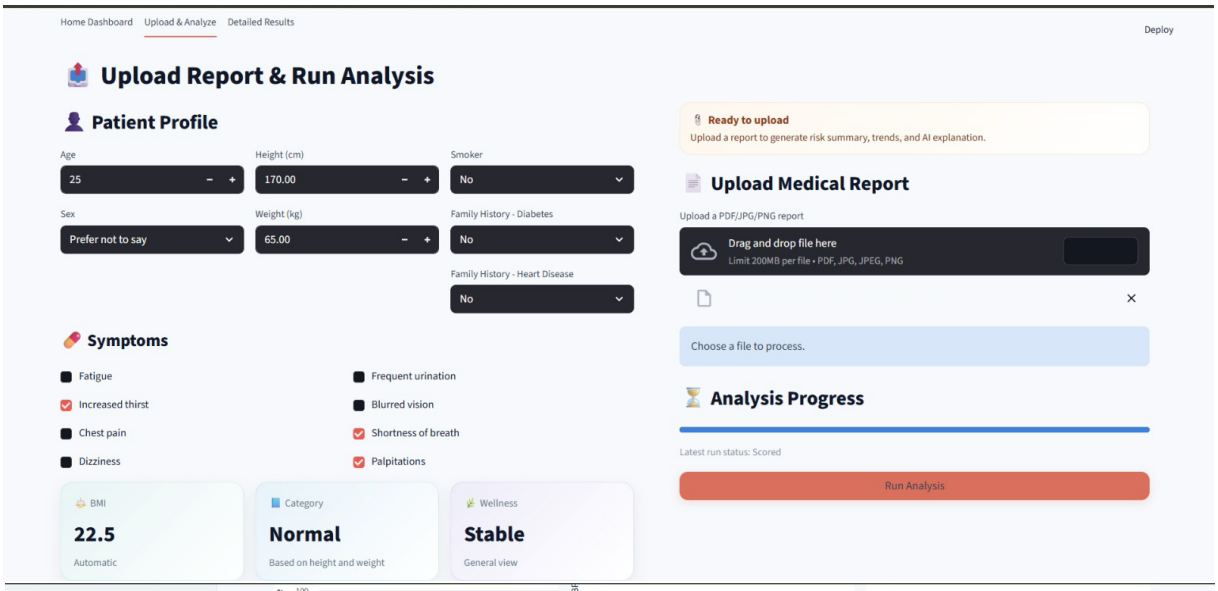
AI Tool	Version	Purpose of Use	Value Added by User
ChatGPT	GPT-5 (Premium)	UI development support and debugging	Adapted generic suggestions into project-specific implementation

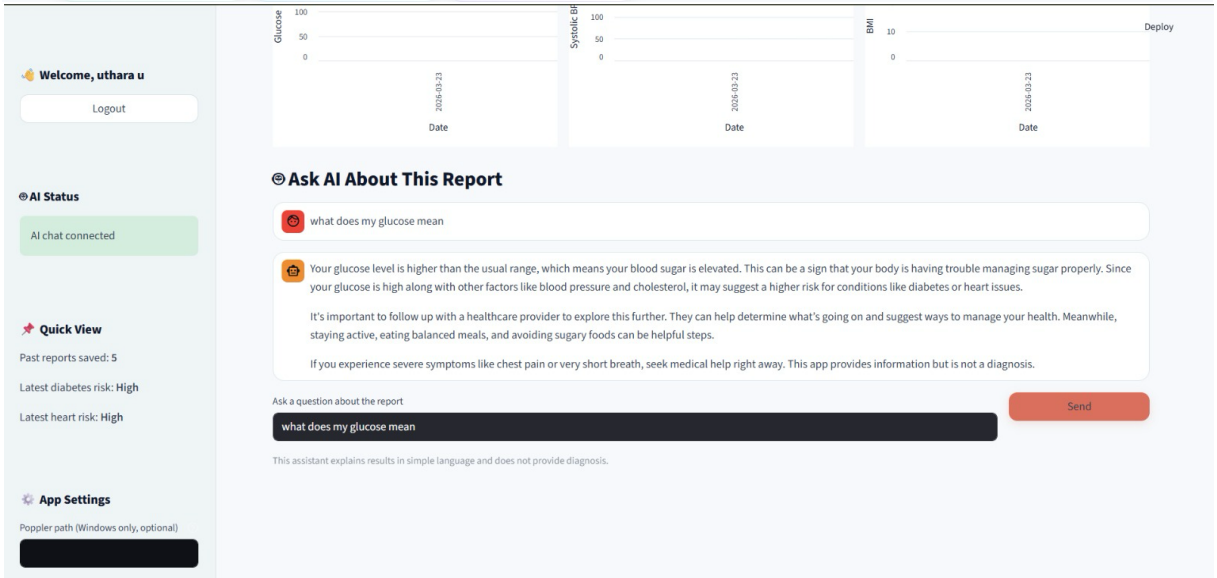
5. APPENDIX

5.1 Appendix A – AI Prompts

1. How to fix UI rendering issues in Streamlit?
2. How to manage historical data for trend visualization?

5.2 Appendix B – Screenshots





Welcome, uthara u

Logout

AI Status

AI chat connected

Quick View

Past reports saved: 5

Latest diabetes risk: High

Latest heart risk: High

App Settings

Poppler path (Windows only, optional)

[redacted]

Key Health Indicators

Deploy

Glucose	Hba1C	Systolic Bp	Diastolic Bp
132.0 mg/dL	6.7 %	145.0 mmHg	92.0 mmHg
High	High	High	High
Reference: 70–99	Reference: < 5.7	Reference: < 120	Reference: < 80

Lifestyle Suggestions

Food
- Choose higher-fiber meals and steadier meal timing.
- Limit salty and heavily processed foods.
- Reduce added sugar and watch portions.

Activity
- Stay active with walking, stretching, or moderate exercise.

Daily Habits
- Track blood pressure regularly.

Follow-Up
- Discuss follow-up blood sugar testing.
- Discuss chest pain or breathing symptoms promptly.

Comparison With Past Report

Glucose	Systolic BP	BMI
— No change	— No change	— No change
Compared with previous report	Compared with previous report	Compared with previous report

Trends Over Time

Home Dashboard Upload & Analyze Detailed Results

Deploy

Detailed Results

Status	Patient ID	Extracted Characters
Scored	AUTO_GEN_2CE6BFBC	1199

Patient Inputs

Age	BMI	BMI Category	Smoker
25	22.5	Normal	No

Extracted Diabetes / Heart Indicators

	Indicator	Value	Unit	Reference	Flag
0	glucose	132	mg/dL	70–99	High
1	hba1c	6.7	%	< 5.7	High
2	cholesterol_total	235	mg/dL	< 200	High
3	ldl	160	mg/dL	< 100	High
4	hdl	38	mg/dL	> 40	Low
5	triglycerides	210	mg/dL	< 150	High
6	systolic_bp	145	mmHg	< 120	High
7	diastolic_bp	92	mmHg	< 80	High

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[redacted]

Deploy

CDSS — Patient Health Portal

Understand your report, key indicators, trends, and next steps in one place.

Home Dashboard Upload & Analyze Detailed Results

Health Dashboard

Overall Health Snapshot

Needs Attention

Some findings may need closer follow-up.

Last analysis status: Scored

Diabetes: High Heart: High BMI: 22.5 Symptoms: 3

Diabetes Risk	Heart Risk	BMI	Symptoms
High	High	22.5	3
ML-based	Rule-based	Normal	Reported

Summary

At a glance

Your report suggests high diabetes risk and high heart risk. This is an informational summary.